

Sresht yyappath

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EDUCATION

BITS Pilani, Hyderabad Campus

B.E. Hons.), Electronics and Communication Engineering

Aug 2020 - Jun 2024

RELEVANT COURSEWORK

Foundations of Data Structures and Algorithms, Object Oriented Programming, Computer Programming, Operating Systems, Digital Design, Communication Networks, Info Theory and Coding, Communication Systems, Digital Signal Processing, Microprocessors and Interfacing

WORK EXPERIENCE

Software Development Engineer Tanla Platforms, Hyderabad

Jun 2024 - Present

Built an Automated Messaging System for Banks: Designed and developed a high-performance system (3000 TPS) using **Java, Spring Boot, and Oracle DB** that fetches message parameters, mobile numbers, and template IDs, identifies matching templates, constructs messages, and sends payloads to the bank's ESB API. Implemented **Redisson-based distributed locking** with **Redis** to prevent duplication across multiple instances. Used **multithreading** to enhance throughput and **Docker** for seamless deployment.

Migrated RCM Platform from RabbitMQ to NATS JetStream: Led proof-of-concept testing and benchmarking to evaluate message brokers, achieving **higher throughput** and reduced latency. Designed and implemented custom components in **Java, Spring Boot, and Apache Camel** for seamless integration with NATS, including **DLQ (Dead Letter Queue)** and **retry mechanisms**, ensuring fault tolerance, reliability, and scalability for high-volume transactional messaging.

Developed Frequency Capping Functionality: Built a **Redis**-based system for Tanla's SMS platform to reduce spam and regulate messaging volume. Implemented daily, weekly, and monthly limits to prevent excessive messaging, ensuring compliance with regulations and improving system reliability, scalability, and overall user experience.

Optimized Template Matching System: Improved template retrieval speed by **15%** through a comparative study of the existing implementation, Elasticsearch, and Lucene-based indexing. Found that **Lucene used less RAM**, making full scans more efficient while maintaining retrieval accuracy and reducing memory overhead.

Intern Hyderabad Consulting Engineers, Hyderabad

Jul 2023 - Dec 2022

Designed 3D electrical substations models for 3 major industrial projects, ensuring compliance with specifications.

Created **single-line diagrams (SLDs)** in AutoCAD for a sugarcane and coal-handling plant, optimizing electrical system planning and ensuring compliance with industry standards.

PROJECTS

Hotel Event Management Web Application

Jan 2023 - May 2023

Designed and built a scalable full-stack hotel event management platform using Angular, Spring Boot, and MySQL, supporting 100+ concurrent bookings with real-time updates, admin controls and role-based user access.

Created an intuitive, responsive UI to streamline event booking and management.

Engineered the back-end with Spring Boot and Java, developing RESTful APIs for seamless communication.

Client-Server Application using POSIX (OS Project)

Feb 2023

Implemented a POSIX-compliant client-server model, enabling real-time request handling and responses.

Used **sockets** to establish efficient communication between client and server.

Optimized request handling by implementing **multi-threading**.

Ensured **fault tolerance** by handling network failures and implementing retry mechanisms.

ACHIEVEMENTS

Rating of 1680 in CodeChef and 1228 in CodeForces.

Rank 39 in CodeChef Division 3 (Rated) and Rank 551 in CodeChef Division 2 (Rated).

Solved 480+ questions on LeetCode.

TECHNICAL SKILLS

C/C++, Java, Python, Javascript, SQL, JSP, SpringBoot, HTML, CSS, React, Linux Terminal, Redis, Nodejs